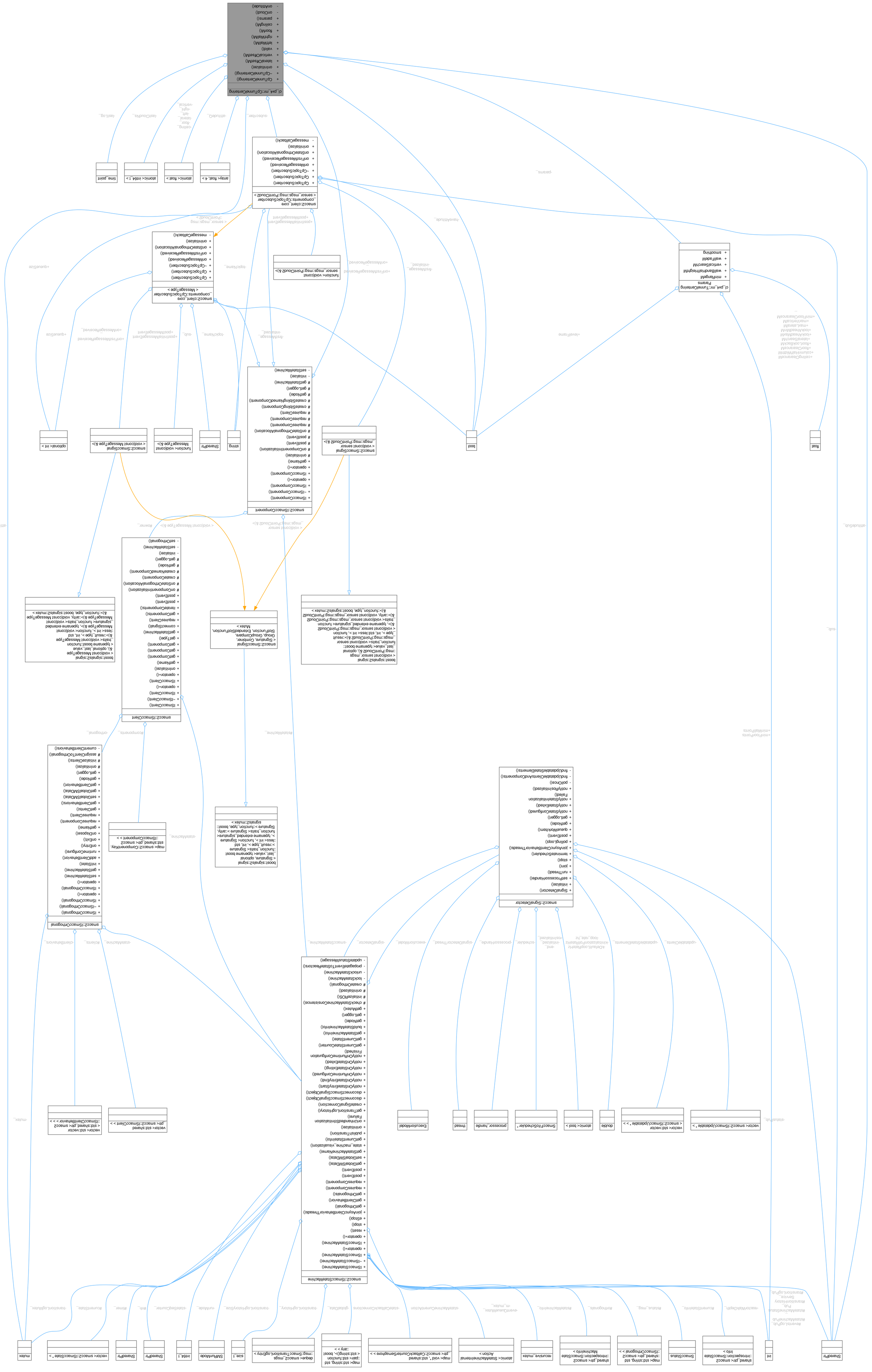
cl_px4_mc::CPTunnelCentering
+ CPTunnelCentering()
+ ~CPTunnelCentering()
+ onInitialize()
+ lateralOffsetM()
+ verticalOffsetM()
+ valid()
+ leftWallM()
+ rightWallM()
+ floorM()
+ ceilingM()
+ params()
- onCloud()
- onAltitude()
-lastLog_
-lastCloudNs_
-ceiling_ -floor_ -lateral_ -left_ -right_ -vertical_
-attitudeQ_
-subscriber_
time_point
atomic< int64_t >
atomic< float >
array< float, 4 >
smacc2::client_core_components::CPTopicSubscriber< sensor_msgs::msg::PointCloud2 >
+ CPTopicSubscriber()
+ CPTopicSubscriber()
+ ~CPTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()
-topicName_
+queueSize_
+onMessageReceived_
+postInitialMessageEvent
+postMessageEvent
< sensor_msgs::msg ::PointCloud2 >
function< void(const sensor_msgs::msg::PointCloud2 &)>
-firstMessage_initialized_
+onFirstMessageReceived_
+haveAttitude_
-params_
cl_px4_mc::TunnelCentering Params
+ minRangeM
+ wallBandHalfHeightM
+ verticalSearchM
+ wallFadeM
+ smoothing
+levelFrame
+ceilingClearanceM
+columnHalfWidthM
+floorClearanceM
+floorLookBackM
+lateralSearchM
+lookAheadMaxM
+lookAheadMinM
+maxLateralM
+maxVerticalM
+minFloorClearanceM
...
-sub_
-attitudeSub_
-attitudeMutex_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ ISmaccComponent()
+ operator=()
+ operator=()
+ setStateMachine()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ iterateComponents()
+ getComponents()
+ postEvent()
+ postEvent()
+ onComponentInitialization()
+ onStateOrthogonalAllocation()
+ createNamedComponent()
+ createComponent()
+ getNode()
+ getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ ISmaccClient()
+ operator=()
+ ISmaccClient()
+ operator=()
+ onInitialize()
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+ connectSignal()
+ requiresClient()
+ iterateComponents()
+ getComponents()
+ postEvent()
+ postEvent()
+ onComponentInitialization()
+ onStateOrthogonalAllocation()
+ createNamedComponent()
+ createComponent()
+ getNode()
+ getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
-topicName_
+queueSize_
+onMessageReceived_
+postInitialMessageEvent
-sub_
-topicName_
-firstMessage_initialized_
+onFirstMessageReceived_
+postMessageEvent
function< void(const sensor_msgs::msg::PointCloud2 &)>
smacc2::client_core_components::CPTopicSubscriber< sensor_msgs::msg::PointCloud2 >
+ CPTopicSubscriber()
+ CPTopicSubscriber()
+ ~CPTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()
-subscriber_
-attitudeQ_
-ceiling_ -floor_ -lateral_ -left_ -right_ -vertical_
-lastCloudNs_
-lastLog_
smacc2::ISmaccOrthogonal
+ ISmaccOrthogonal()
+ ~ISmaccOrthogonal()
+ ISmaccOrthogonal()
+ operator=()
+ ISmaccOrthogonal()
+ operator=()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runTimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
-stateMachine_
-clients_
-clientBehaviors_
-orthogonal_
#components_
#stateMachine_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ ISmaccStateMachine()
+ operator=()
+ ISmaccStateMachine()
+ operator=()
+ resetI()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ onUnhandledInitialization Failure()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryEnd()
+ notifyOnStateEntryStart()
+ notifyOnRuntimeConfigured()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
-smaccStateMachine_
-signalDetector_
-executionModel_
-signalDetectorThread_
-processorHandle_
-scheduler_
-end_initialized_
-rosInitialized_
-kDefaultLoopRateHz
-kInitializationPollRateHz
-loop_rate_hz
-updateClients_
-updateableStateElements_
-updateableClients_
-attitudeMutex_
-mutex_
-clientBehaviors_
-clients_
-stateMachine_
-orthogonal_
#components_
#stateMachine_
vector< std::shared_ptr< smacc2 ::ISmaccClient > >
vector< std::shared_ptr< smacc2 ::ISmaccClient > >
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
-stateMachine_
smacc2::SmaccSignal< void(const sensor_msgs::msg::PointCloud2 &)>
boost::signals2::signal< SignalType, optional_last_value< typename boost::result_of< Combiner(InputIterator, InputIterator)>::type >, int, std::less< int >, function< void(const Message1Type &, const Message2Type &)> >
boost::signals2::signal< void(const sensor_msgs::msg::PointCloud2 &), optional_last_value< typename boost::result_of< Combiner(InputIterator, InputIterator)>::type >, int, std::less< int >, function< void(const sensor_msgs::msg::PointCloud2 &)>, traits< void(const sensor_msgs::msg::PointCloud2 &)>, typename extended_signature< function_traits< void(const sensor_msgs::msg::PointCloud2 &)>::function_type, boost::signals2::mutex >
-transitionLogMutex_
-currentState_
-hm_
-time_
-stateSeqCounter_
-runMode_
-transitionLogHistorySize_
-transitionLogHistory_
-globalData_
-stateCallbackConnections_
-stateMachineCurrentAction_
-eventQueueMutex_
-m_mutex_
#stateMachineInfo_
#orthogonals_
#status_msg_
#currentStateInfo_
#reactorWalkDepth_
#stateMachineStatusPub_
#transitionHistoryService_
#transitionLogPub_
#eventLogPub_
#stateMachinePub_
mutex
vector< smacc2::ISmaccState * >
SharedPtr
SharedPtr
int64_t
SMHunMode
size_t
deque< smacc2_msgs ::msg::SmaccTransitionLogEntry >
map< std::string, std::function< std::string(), boost::any > >
atomic< StateMachineInternalAction >
recursive_mutex
shared_ptr< smacc2 ::introspection::SmaccStateMachineInfo >
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
SmaccStatus
shared_ptr< smacc2 ::introspection::SmaccState Info >
int
SharedPtr
-mutex_
vector< std::vector< smacc2::ISmaccClient * > >
vector< std::shared_ptr< smacc2 ::ISmaccClient > >
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ ISmaccStateMachine()
+ operator=()
+ operator=()
+ resetI()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ onUnhandledInitialization Failure()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryEnd()
+ notifyOnStateEntryStart()
+ notifyOnRuntimeConfigured()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
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# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
-smaccStateMachine_
-signalDetector_
-executionModel_
-signalDetectorThread_
-processorHandle_
-scheduler_
-end_initialized_
-rosInitialized_
-kDefaultLoopRateHz
-kInitializationPollRateHz
-loop_rate_hz
-updateClients_
-updateableStateElements_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ stop()
+ joinAsyncClientBehaviorThreads()
+ terminateScheduler()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyStateInitialization Failed()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
-params_
-sub_
-attitudeSub_
vector< smacc2::ISmaccUpdatable * >
vector< std::vector< smacc2::ISmaccUpdatable * > >
double
atomic< bool >
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
-statusPub_
+minFloorPoints
+minWallPoints
float
bool
smacc2::SmaccSignal< void(const sensor_msgs::msg::PointCloud2 &)>
boost::signals2::signal< SignalType, optional_last_value< typename boost::function_traits< SignalType >::function_type, boost::signals2::mutex >
-stateMachine_
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
-stateMachine_
-components_
-orthogonal_
vector< std::shared_ptr< smacc2 ::ISmaccClient > >
vector< std::vector< smacc2::ISmaccClient * > >
-mutex_
smacc2::ISmaccClient
+ ISmaccClient()
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+ createNamedComponent()
+ createComponent()
+ getNode()
+ getLogger()
- initialize()
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- setOrthogonal()
-stateMachine_
-clients_
-clientBehaviors_
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vector< std::shared_ptr< smacc2 ::ISmaccClient > >
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